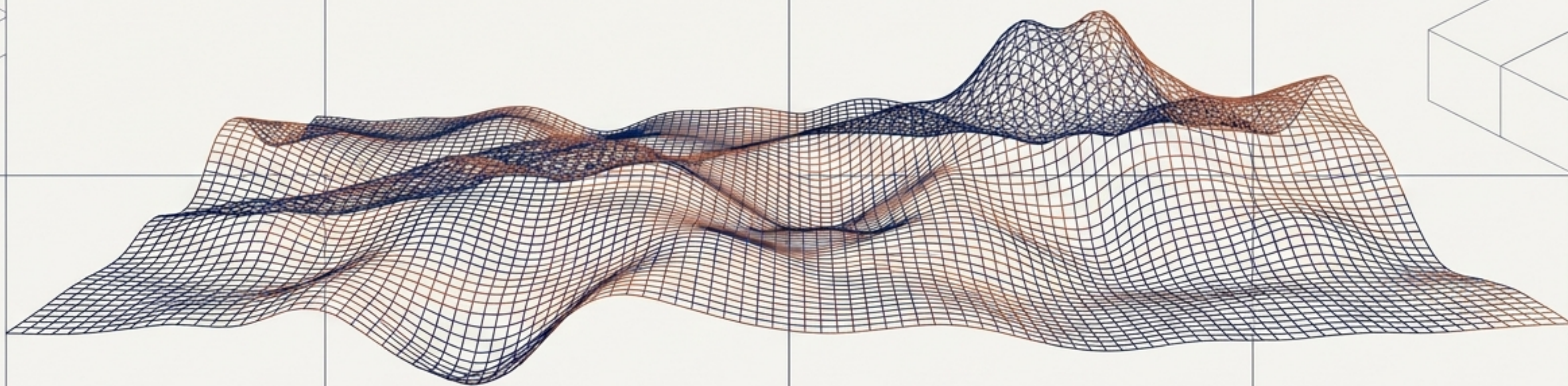




Symm: Algorithmic Trading Architecture

Decoding the Tick-to-Trade Pipeline & 3D Telemetry Dashboard

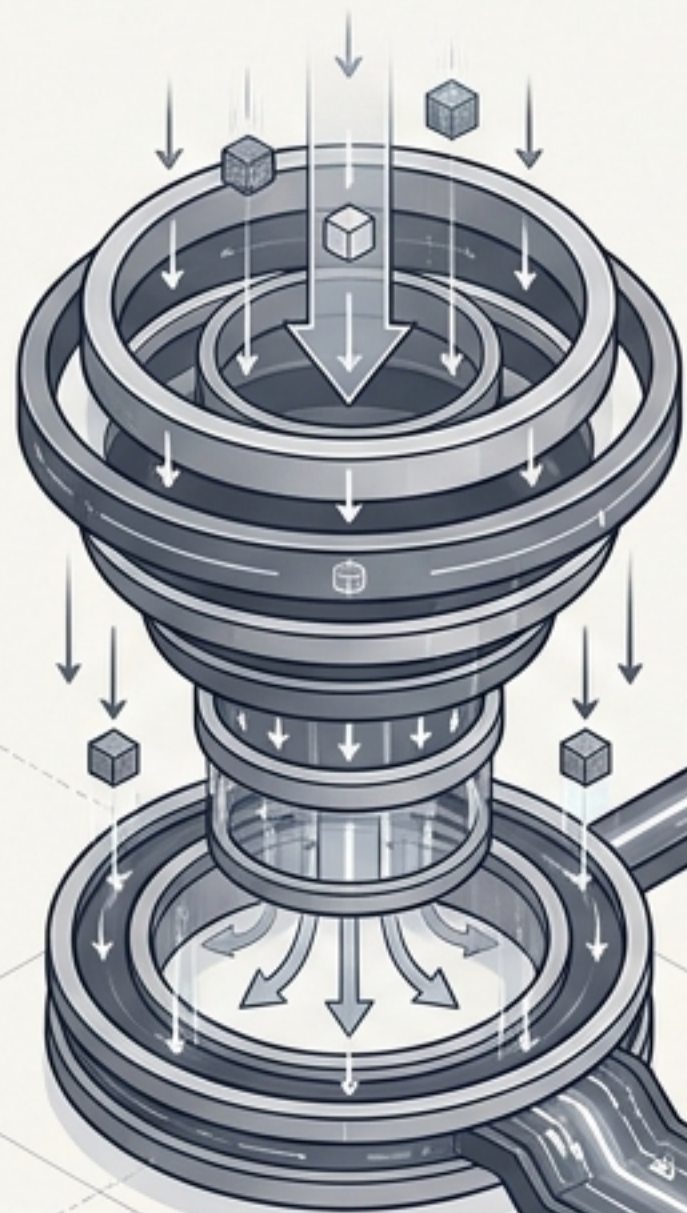
Engine: Go (Backend) /
Optimizer (MCTS)

Dashboard: React /
TypeScript / SciChart 3D

Exchange: Kraken
WebSockets (L2/L3)

The Exchange Layer (Ingestion)

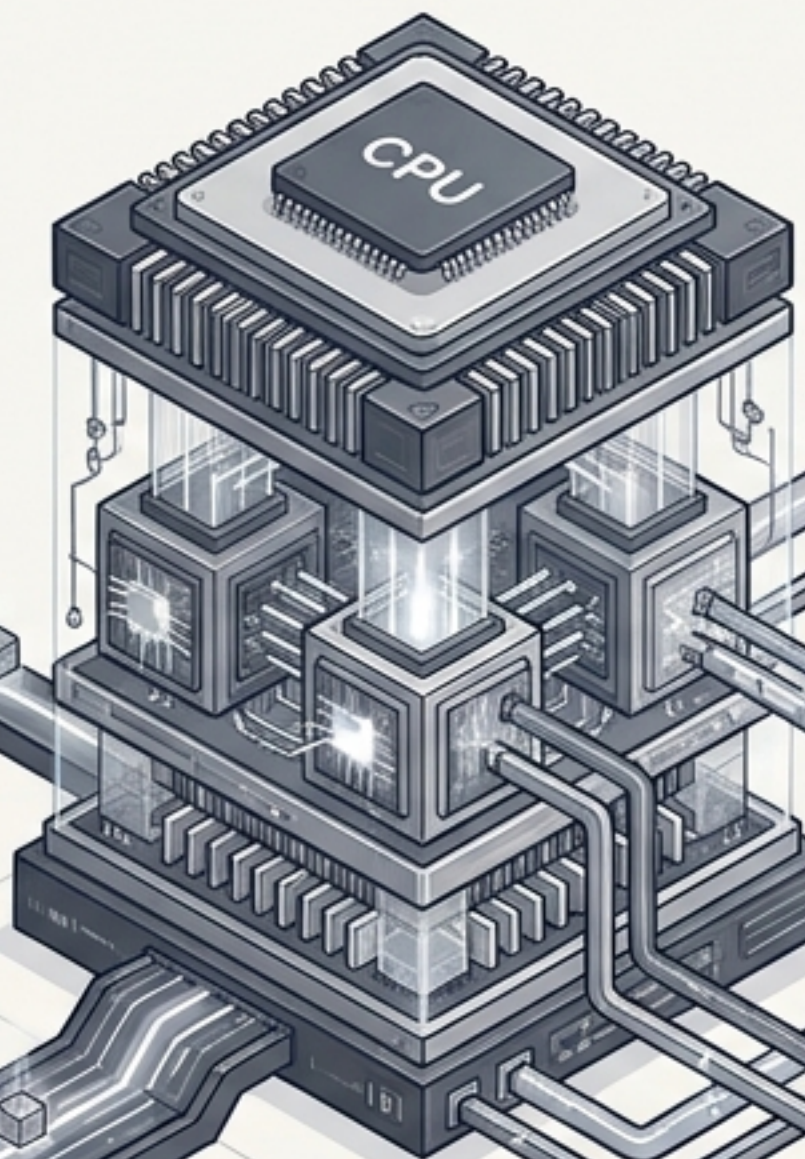
public
private
paper
replay



High-throughput WebSocket v2 connections pulling L2/L3 order books, 24h tickers, and trade tape.

The Core Engine (Intelligence)

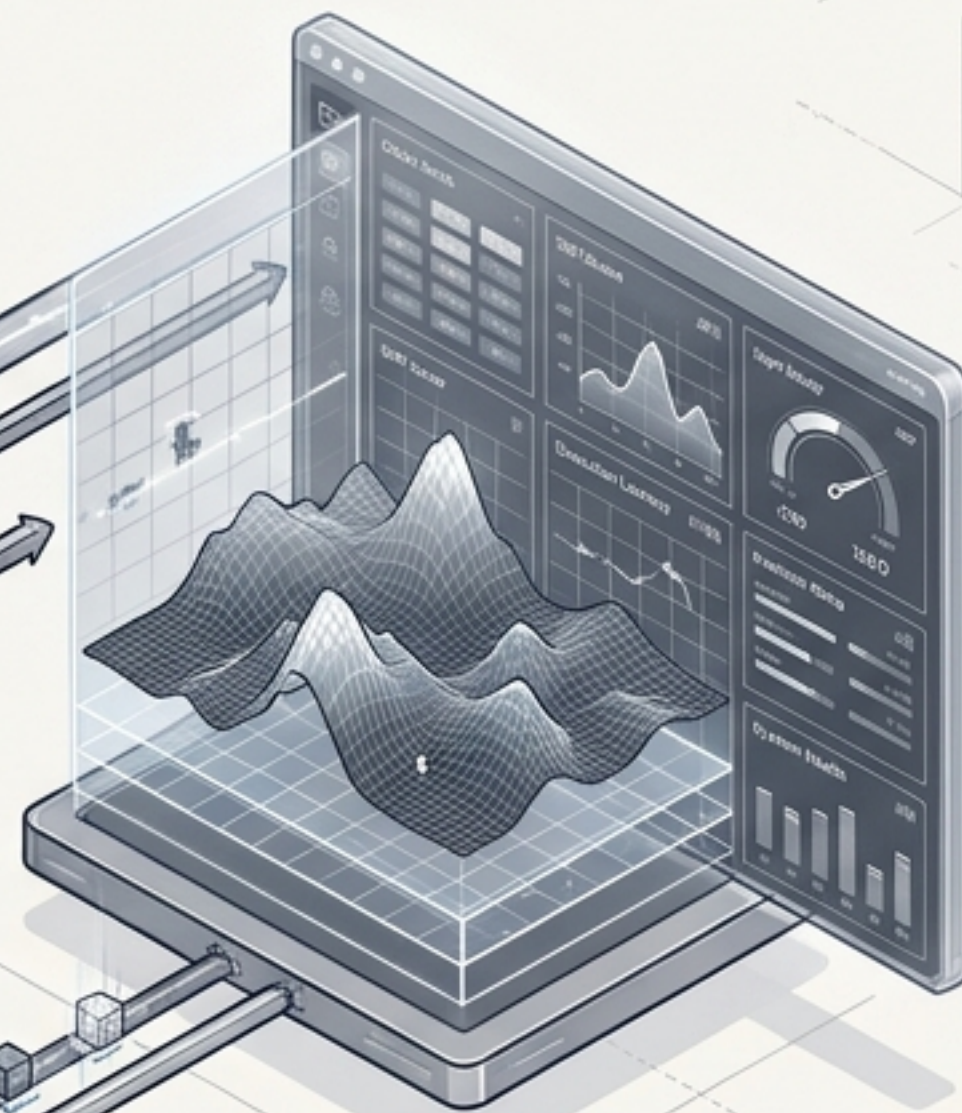
evaluator.go optimizer



The logic core. Extracts Signal-to-Noise Ratio (SNR), evaluates decision trees, and executes trades.

Telemetry Dashboard (Observability)

DashboardLayout.tsx ws-stream.ts



Real-time 3D topographical map and signal gauges routing Go events into a visual terminal.

DATA INGESTION & ROUTING

Public Market Data

REST (/public/AssetPairs, /public/OHLC)

WebSockets (book, ticker, trade)

Pushes snapshots and incremental updates (e.g., BookLevel with 10-level CRC32 checksums).

Private Execution Data

Authenticated WebSockets (TokenProvider credentials)

Ingests L3 order events (OrderEvent), execution fills (ExecutionFee), and user balances (BalanceWallet).

The qpool.Q Broadcast

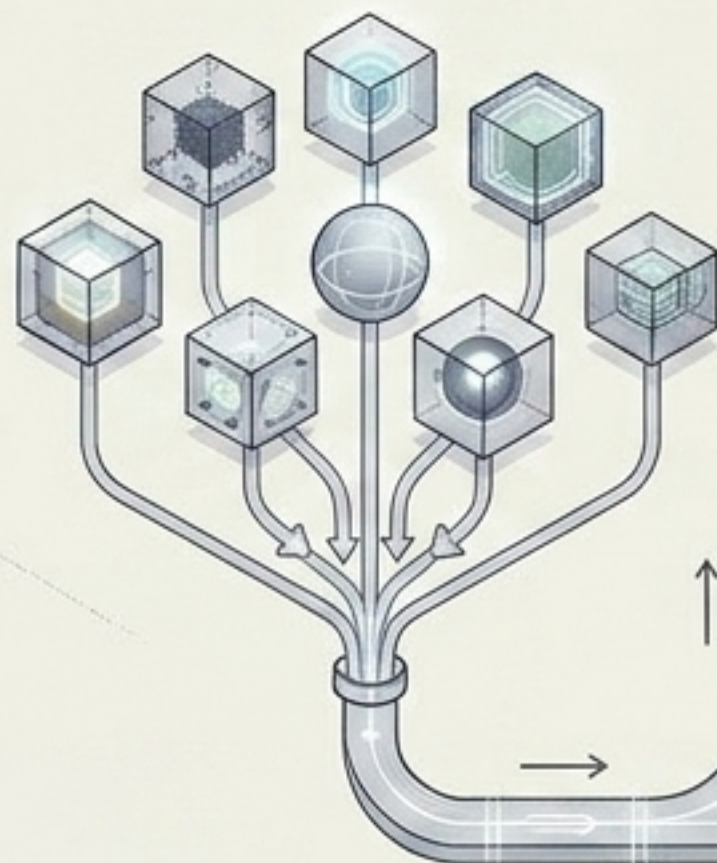
Shared Memory Bus. All streams decode into structured Go structs and multiplex to subscribers.

TRADING EXECUTION: LIVE VS. PAPER

	Live Trading (kraken/trading)	Paper Trading (kraken/paper)
Order Matching	Submits <code>trading.AddParams`</code> directly to Kraken API.	Internal matching engine (<code>openOrder`</code> map), catching Limit and Market orders.
Price Execution & Slippage	Filled strictly by exchange order book liquidity.	Simulated execution using <code>halfSpreadBps`</code> (2.5) and <code>marketSlippageBps`</code> (1.0) penalties.
Balance Tracking	Streams real wallet updates from Kraken <code>user.Balance`</code> .	Emulates <code>Balances`</code> object, deducting synthesized fee and managing assets map.
Identifier Minting	Tracks native <code>OrderID`</code> and <code>ClOrdID`</code> .	Identifier struct mints hex-based dummy IDs (<code>PAPER-`</code> , <code>p`</code>).

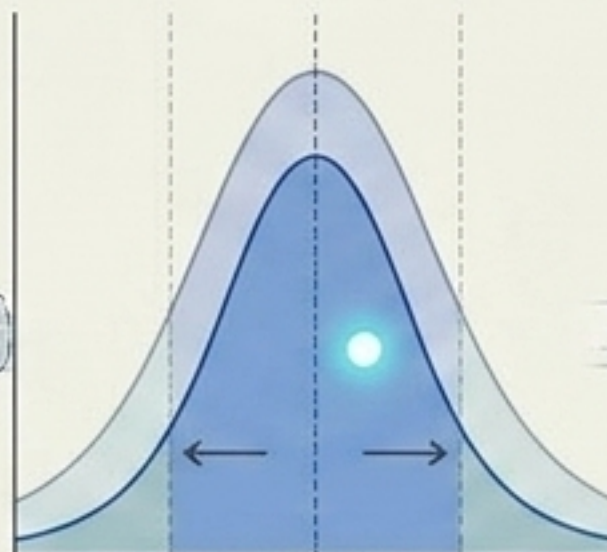
DATA TRANSFORMATION FUNNEL

1. Raw Fused Strength



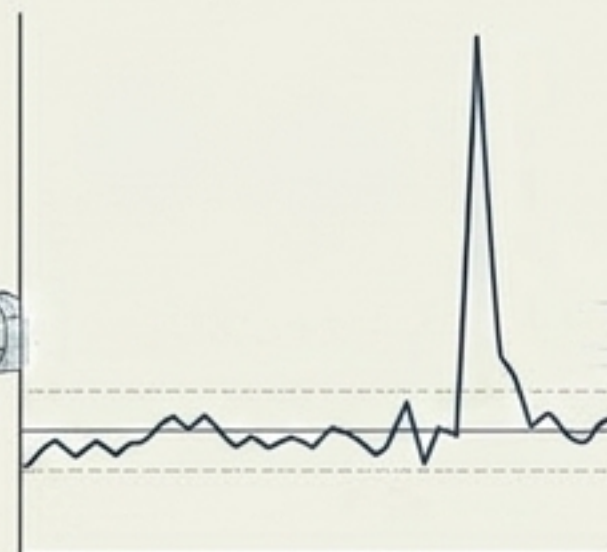
Data points from 8 distinct sources (hawkes, fluid, pumpdump, depthflow, etc.) are synthesized into a baseline strength value.

2. Confidence (Band Clarity)

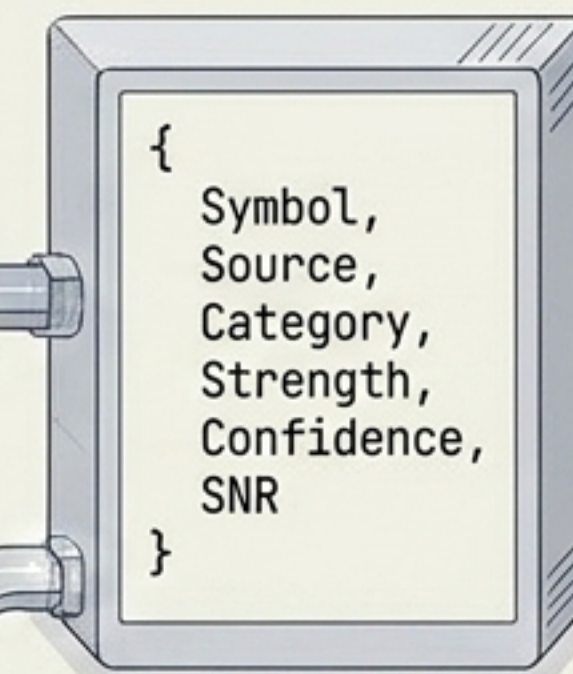


Cross-sectional clarity.
Measures how decisively a reading sits inside its assigned category band (margin to the boundary).

3. Signal-to-Noise Ratio (SNR)

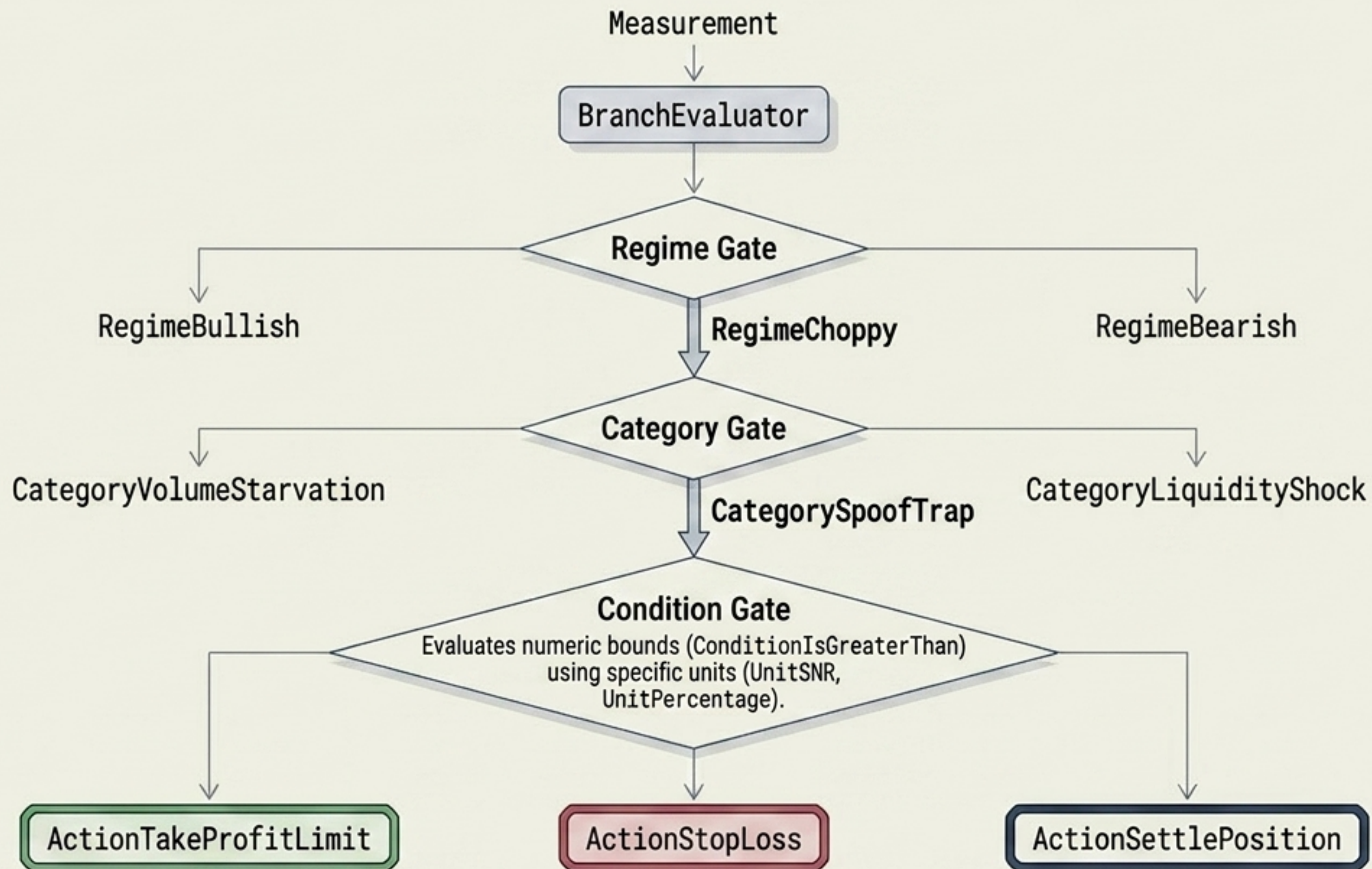


Temporal surprise.
Measures how many standard deviations the current Confidence sits above the symbol's recent baseline.

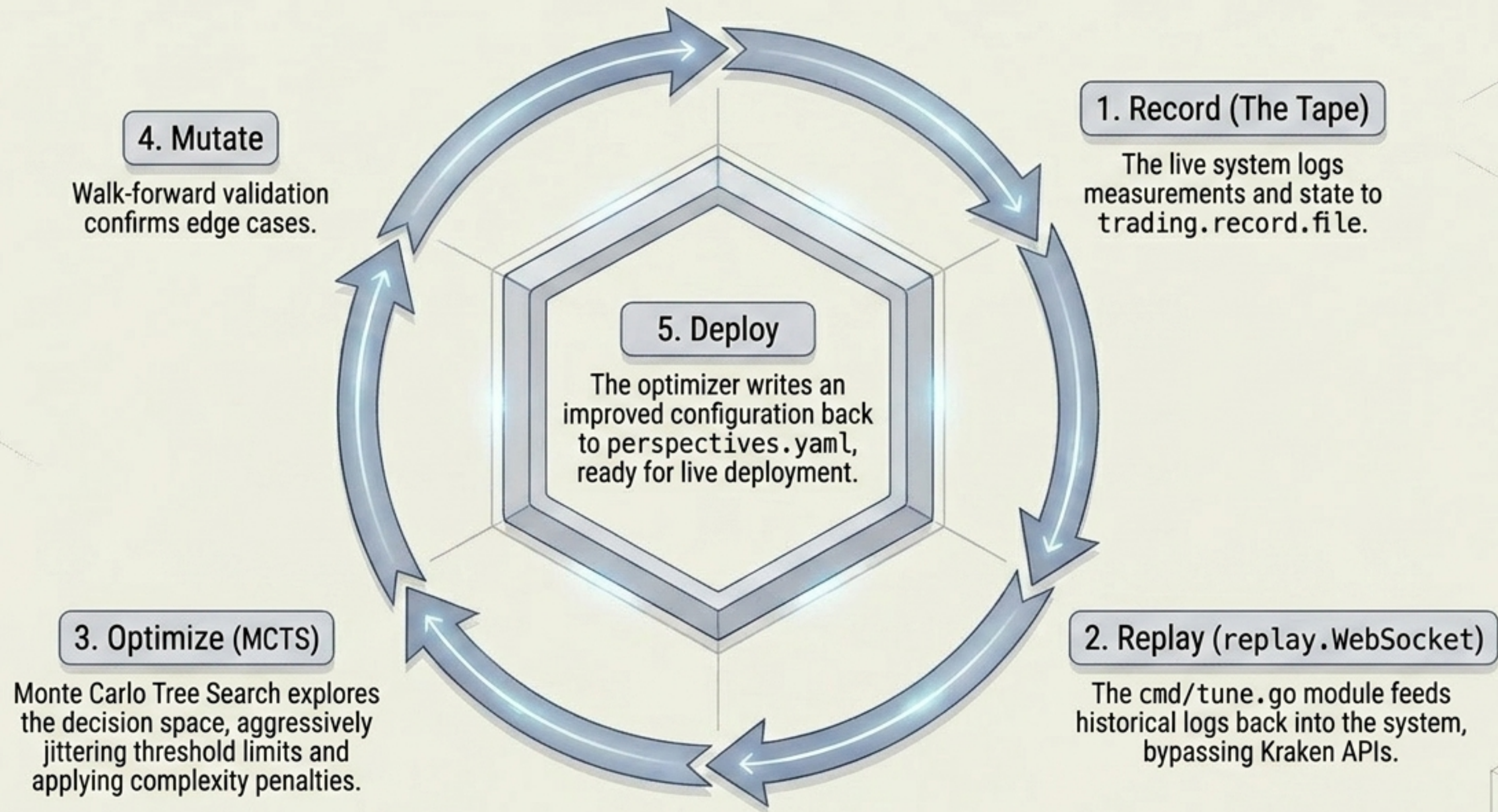


Insight: The engine does not trade on price; it trades on the sudden emergence of clarity out of historical noise.

BRANCH EVALUATOR LOGIC



QUANT FLYWHEEL

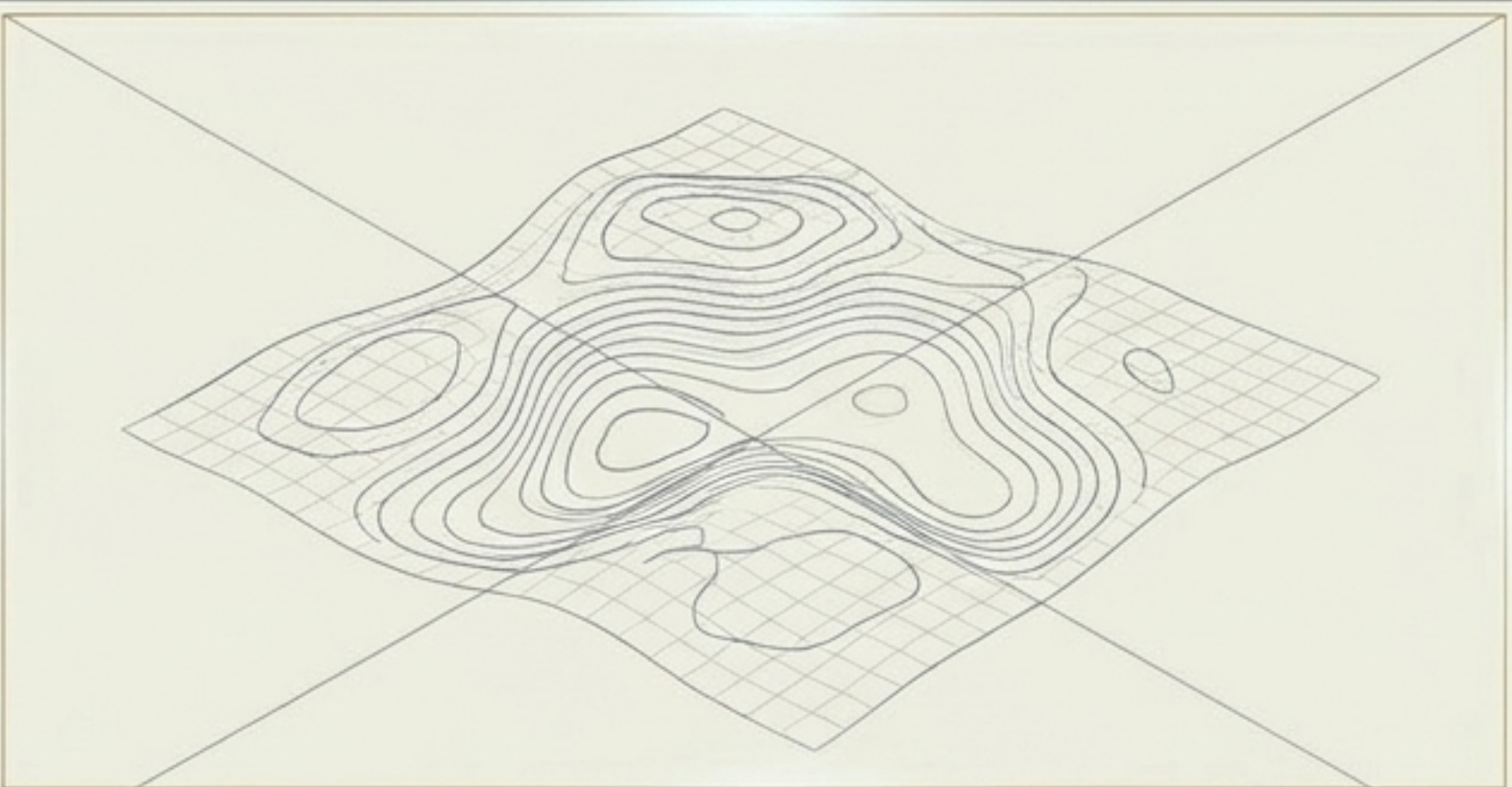


INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: DASHBOARD UI LAYOUT SCHEMATIC

DashboardLayout.tsx

DashboardHeader.tsx

Tracks wallet balance, reserved EUR, and engine pulse status.



FluidSurfaceChart.tsx
The 3D Topology view (field_snapshot stream).



Gauges.tsx & SignalHeatmap.tsx
4-sigma SNR monitoring.

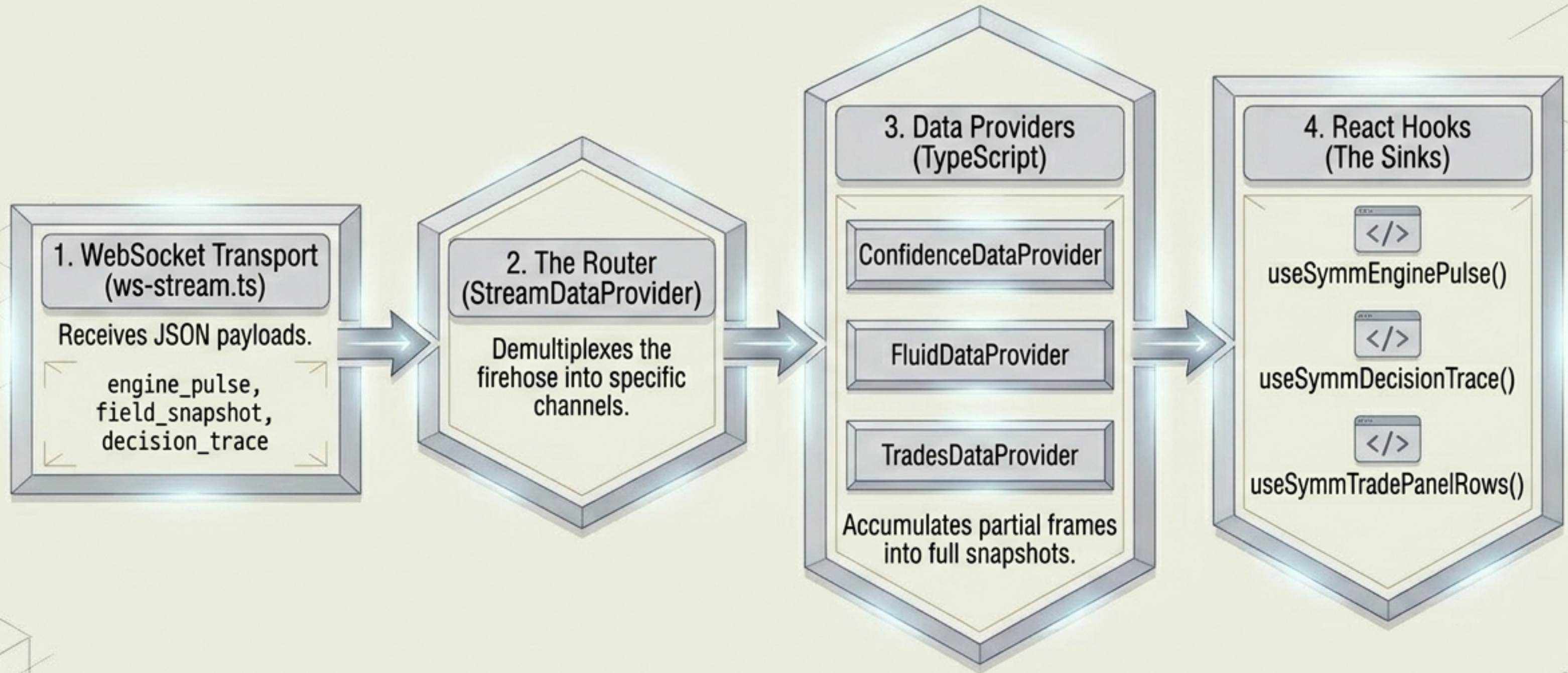


TradeChart.tsx & PredictionChart.tsx
Financial charting and forward vectors.

ID	Type	Asset	Amount	Price	Status	Time	Details

Decisions.tsx & Trades.tsx
Audit tables and execution tracking (wallet & candle_bar streams).

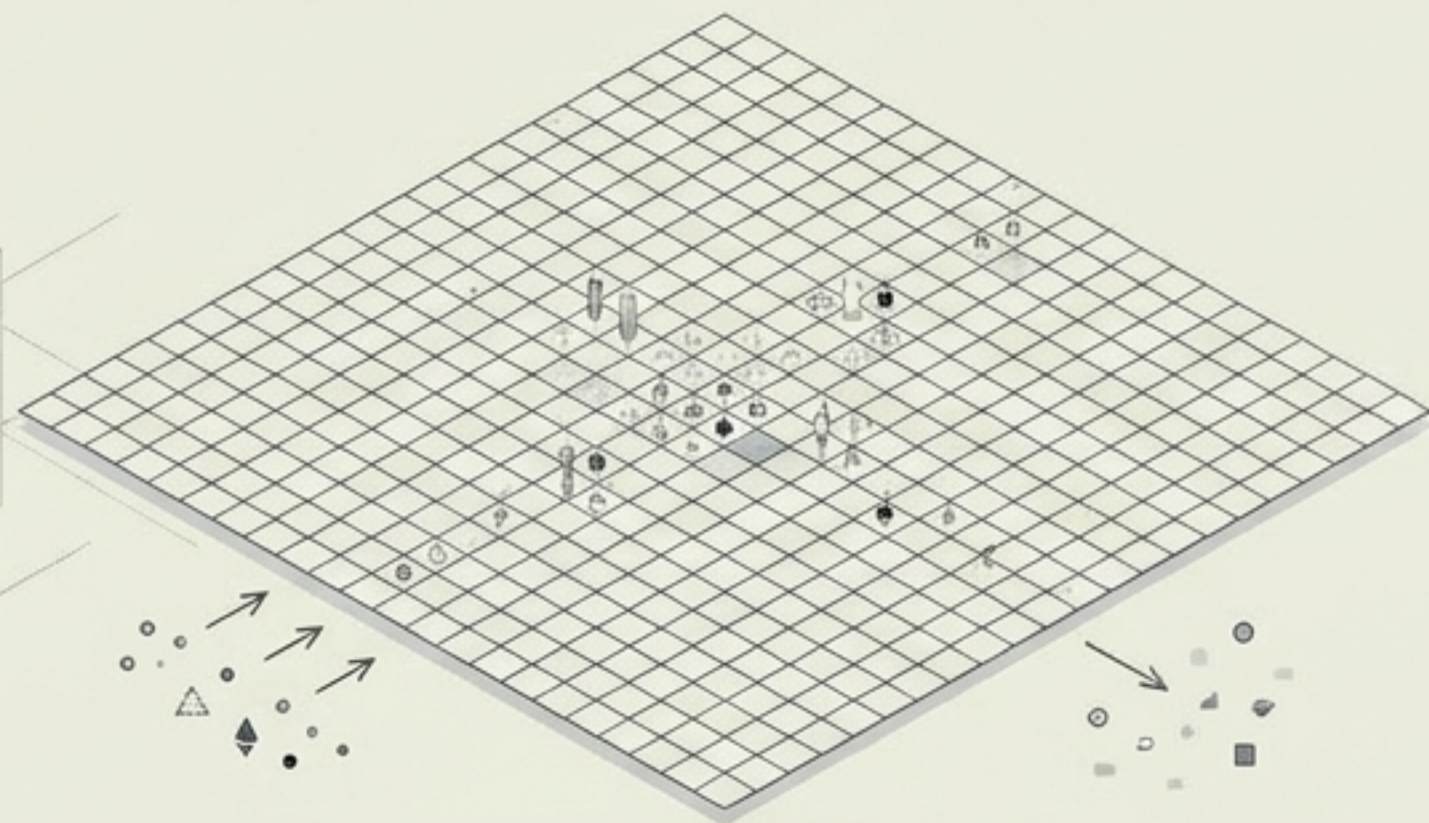
INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: DATA PIPELINE SCHEMATIC



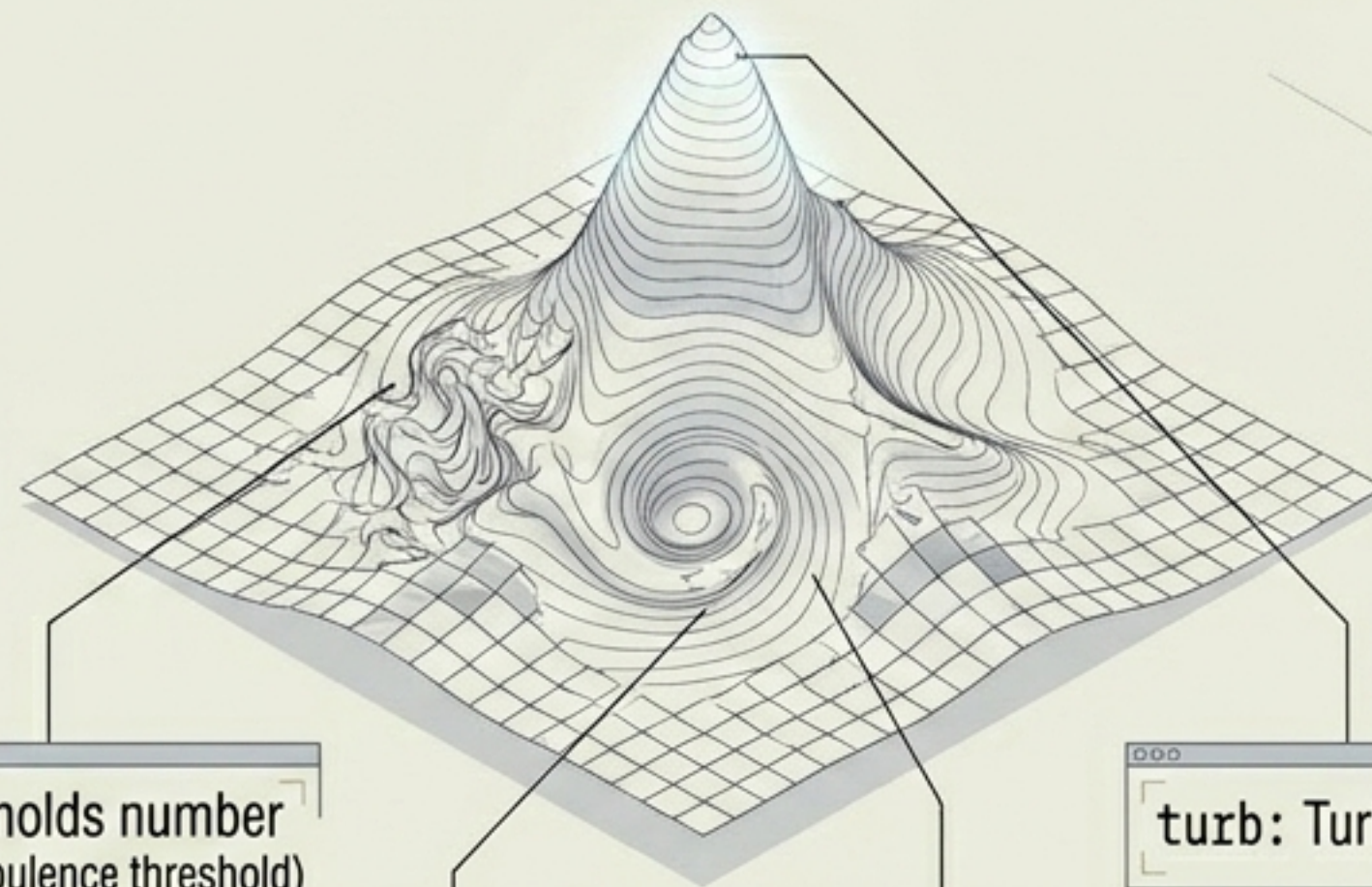
INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: 2D DATA TO 3D TOPOGRAPHY TRANSLATION

The Grid (X/Y Axis)

The Lift (Z-Axis)



Symbols are binned on a 32x32 grid based on their change percentage and volume rank.



re: Reynolds number
(market turbulence threshold)

turb: Turbulence

A symbol's height on the map is not price—it is determined by its Fluid Activity.

vort: Vorticity
(rotational price action)

div: Divergence

Result: The algorithm identifies hotspots in the market by looking for towering peaks of fluid activity, trading topological anomalies rather than price lines.

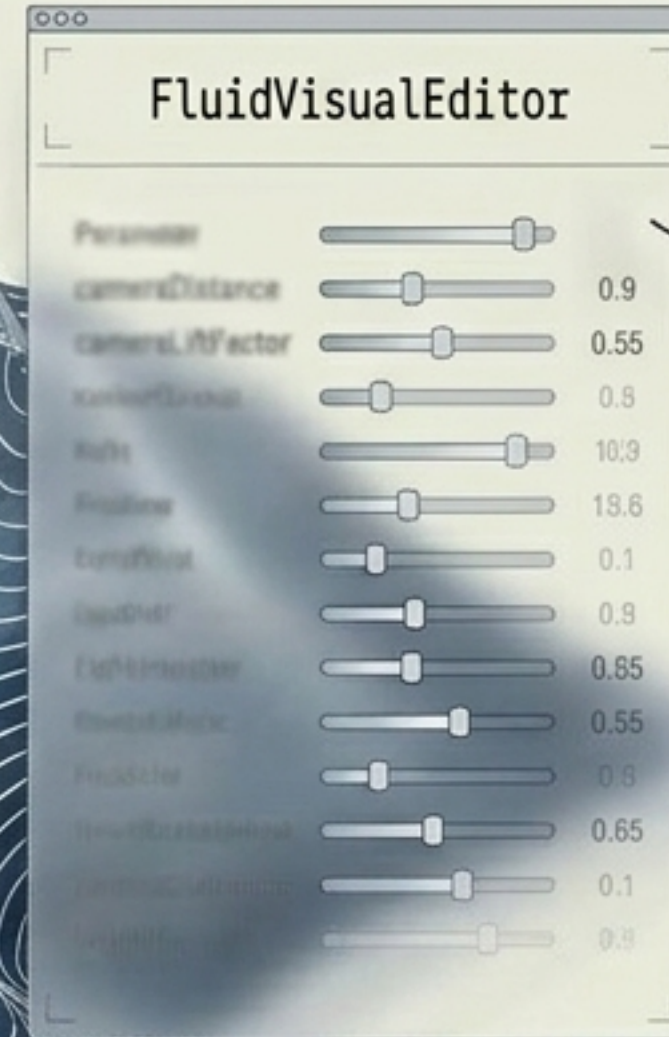
INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: VISUAL RENDERING & SMOOTHING

① EMA Smoothing ($\alpha = 0.35$)

Softens isolated spikes over time to track persistent hotspots.

② Spatial Smoothing

Gaussian blur (`smoothHeightmapSpatial`) lifts neighboring cells to group **correlated** market moves.



③ Camera Controls

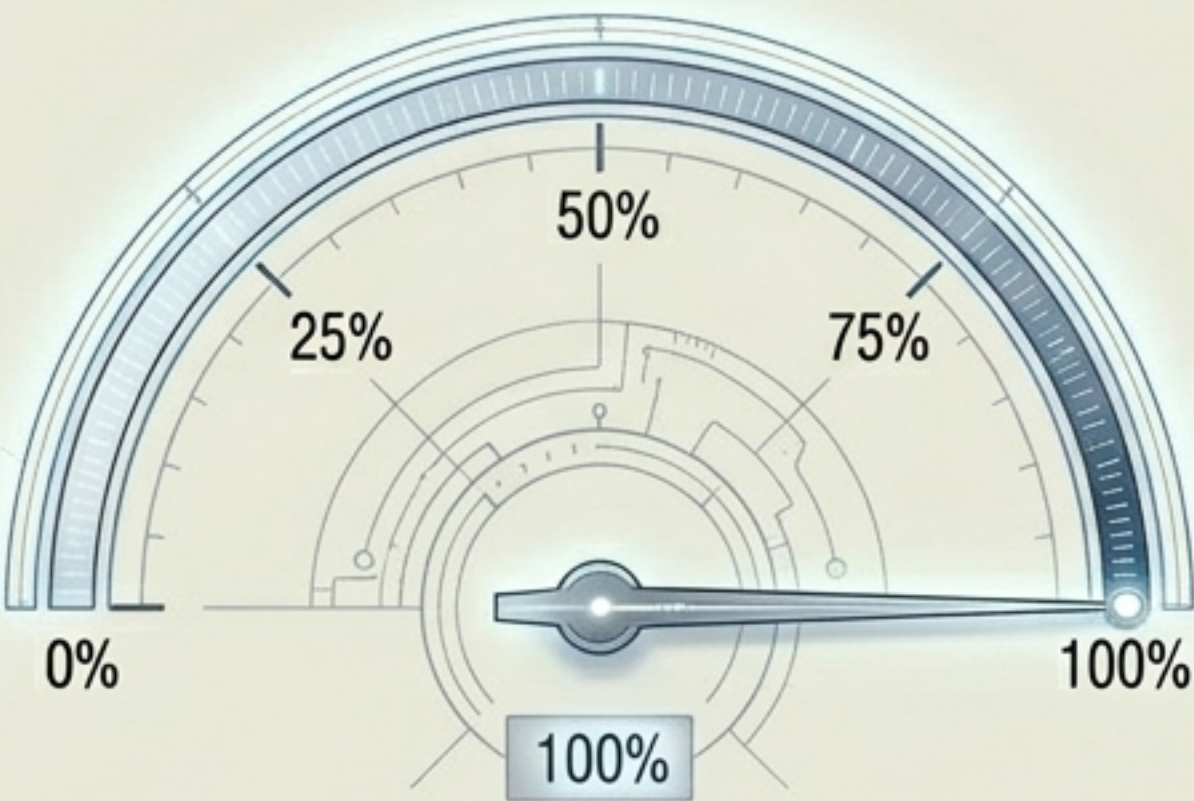
`cameraDistanceFactor` (0.9) and `cameraLiftFactor` (0.55) adjust the isometric viewing angle.

④ Visual Enhancements

Configurable shininess, `lightingFactor`, and `contourStrokeThickness` emphasize terrain gradients.

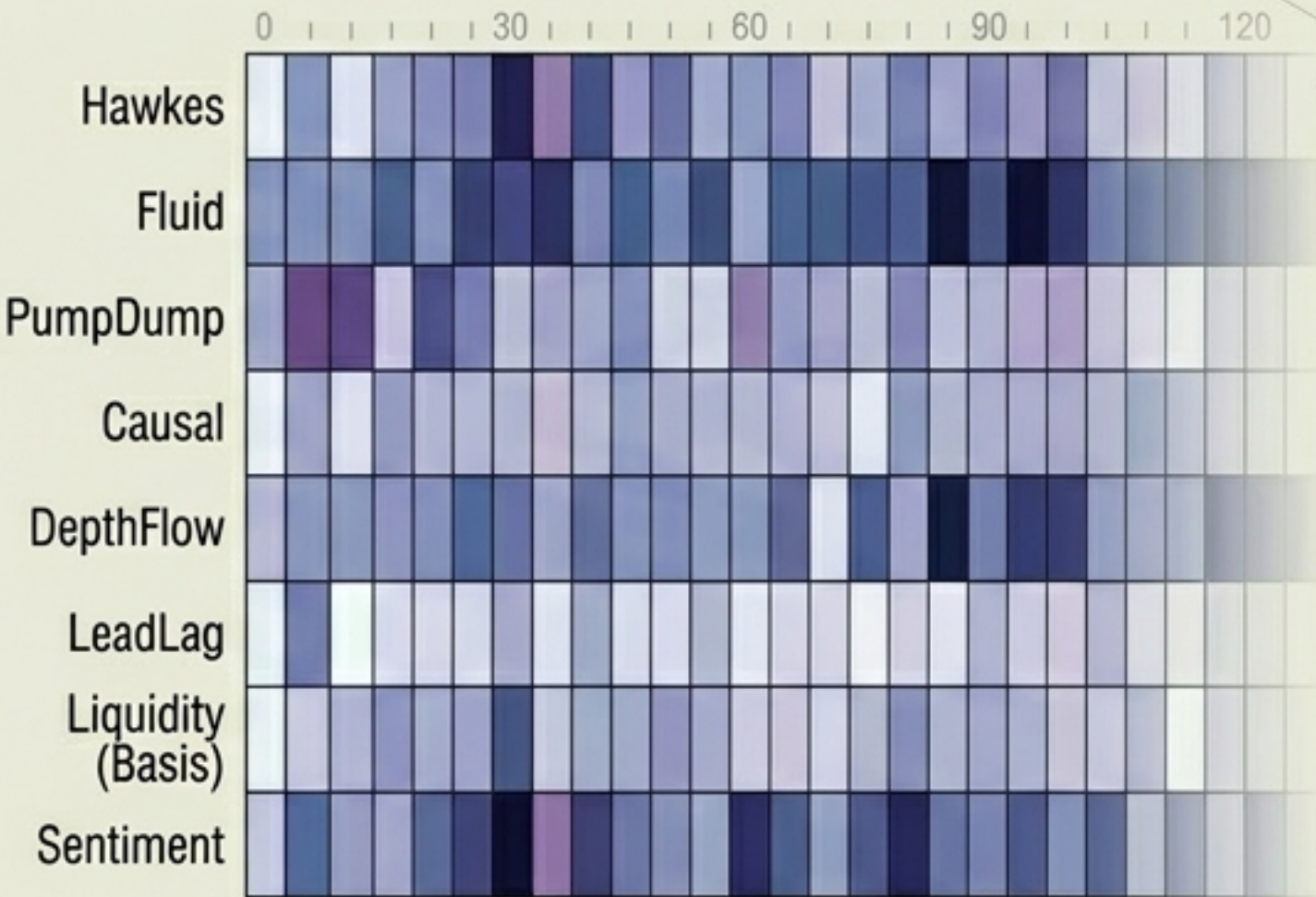
INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: UI COMPONENT DETAIL

The Signal Gauge (draw-signal-gauge.ts)



Tracks from 0 to 100%. The scale represents a 4-sigma Signal-to-Noise Ratio limit (GAUGE_FULL_SIGMA = 4). A reading sits at the noise floor around 25% of the dial, and pins at 100% only during strong, unprecedented statistical spikes.

The Rolling Heatmap (SignalHeatmap.tsx)



Tracks the instantaneous confidence of 8 distinct intelligence signals over time.

INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: OBSERVABILITY & DECISION AUDIT

Trade Panel (TradesPanel)		
Position	Metrics	Unrealized
BTC/EUR Long	Size: 2.5 BTC, Mark: €38,500.25, Entry: €37,950.10	+ €1,375.38
ETH/EUR Short	Size: 40 ETH, Mark: €2,850.75, Entry: €2,920.30	+ €2,782.00
SOL/EUR Long	Size: 150 SOL, Mark: €98.10, Entry: €105.45	- €1,102.50

Highlights Unrealized EUR
with distinct formatting
(formatSignedEur, pnlTone)

Mark Price

Entry Price

Audit & Decision Table (DecisionTable)				
Symbol	Condition	Verdict Badge		Why Code
BTC/EUR	Signal > Threshold (3.0)	✓ ALLOW	Trade condition met.	signal_strength_v1
ETH/EUR	Signal in Range (1.5 - 2.5)	● BELOW	Signal in play, but below threshold.	signal_muted
SOL/EUR	Volume Starvation (< 5%)	✗ DENY	Gate rejected the logic.	volume_starvation
ADA/EUR	Execution Risk (High)	✗ DENY	Gate rejected the logic.	risk_gate_high

Why codes (e.g., volume_starvation) map directly to the Perspectives logic.

INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: FORECAST & ERROR TRACKING



A visual representation of the engine continuously correcting its own thesis. It plots realized cross-sections versus forward forecasts, tracking the mathematical gap between expectations and reality.

INSTITUTIONAL QUANT ARCHITECTURE BLUEPRINT: SYMM CLOSED-LOOP ENVIRONMENT

